

IMAGE ENHANCEMENT USING PIXEL-BASED AND SPATIAL FILTERING METHODS

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Introduction

Image enhancement is basically about making an image easier to see & understand. Different images can have different problems, so obviously 1 enhancement method will not work perfectly for everything. For this assignment, I tested 4 different image problems: blur, excessive brightness, darkness, and low contrast. Each problem was created from a different original image, then I applied a method that matched the problem. I used **sharpening** for the blurred image, **gamma correction** for the bright image, **logarithmic transformation** for the dark image, and **contrast stretching** for the low-contrast image. After that, I compared the images before & after enhancement visually and also used PSNR (Peak Signal-to-Noise Ratio) to see how close the results were to their original reference images!

Methodology

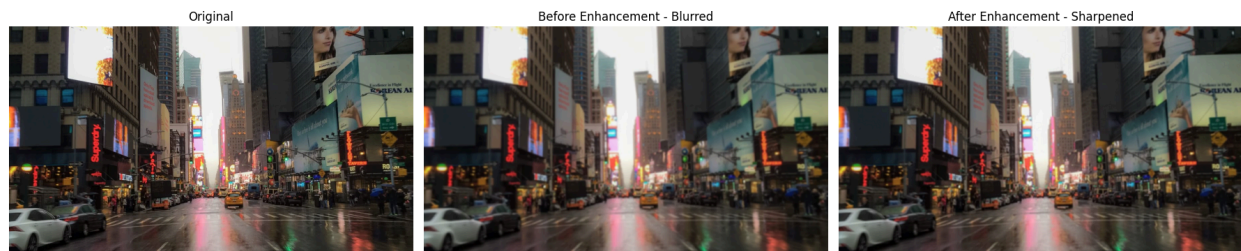
For this experiment, I used 4 original images as the reference images. Instead of finding images that were already damaged, I intentionally created one problem for each image. This way, I could keep the clean original image and use it as a reference when calculating PSNR. each image was given a different problem and then enhanced using a suitable method:

Image problem	How i created the problem	Enhancement method
Blur	Gaussian blur	Sharpening filter
Bright	Increased the luminance	Gamma correction
Dark	Reduced the luminance	Logarithmic transformation
Low contrast	Reduced the intensity range	Contrast stretching

The original image, degraded image, & enhanced image were then compared visually. I also calculated the PSNR before and after enhancement to check whether the enhancement made the image closer to the original reference.

Results and analysis

Blurred image



For the first experiment, I intentionally blurred the image using gaussian blur. The difference is pretty noticeable, especially around the signs, buildings, vehicles, and other small details. After applying

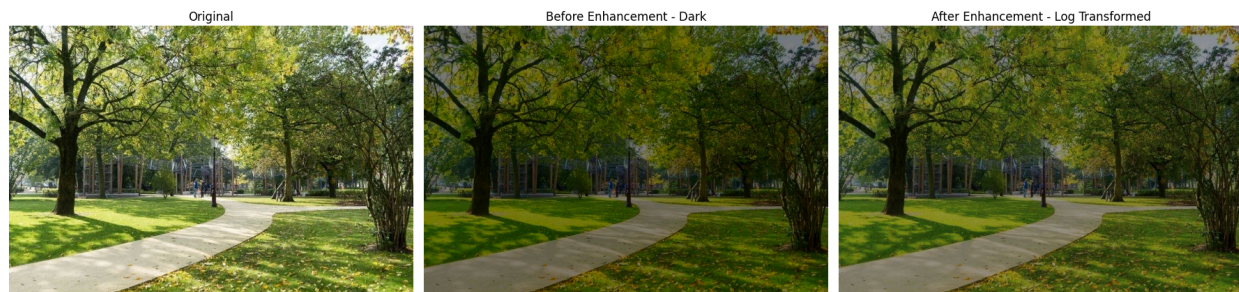
the sharpening filter, the edges became clearer again! The PSNR increased from 26.88 dB to 27.89 dB, giving an improvement of 1.01 dB. The improvement is not huge, but this makes sense because sharpening mainly emphasizes the edges that are still available. It can't fully bring back details that were already lost during the blurring process. Still, both the visual result & PSNR show that the sharpened image is closer to the original than the blurred version.

Bright image



For the 2nd experiment, I increased the luminance of the image until it became overly bright. This made some parts of the image look washed out & reduced the visibility of details, especially around the road, trees, vehicles, and buildings. After applying gamma correction, the brightness became much more balanced & the image looked closer to the original. The PSNR increased from 16.13 dB to 23.51 dB, which is an improvement of 7.38 dB. Compared to the blurred image experiment, the improvement here is much more noticeable both visually & from the PSNR value.

Dark image



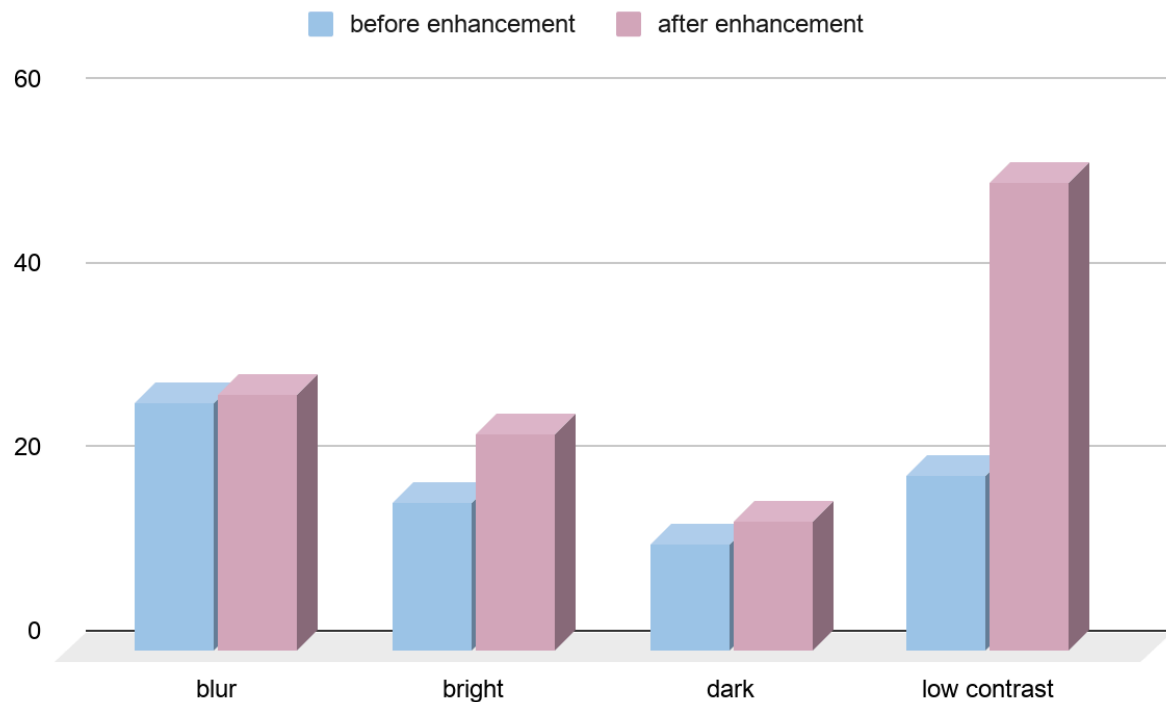
For the 3rd experiment, I reduced the luminance to make the image intentionally dark. As expected, a lot of details became harder to see, especially around the trees, grass, and pathway. After applying the logarithmic transformation, the darker areas became more visible again. The enhanced image did not completely return to the same brightness as the original, but that's expected because the transformation is mainly used to expand darker intensity values rather than directly reverse the darkening process. The PSNR increased from 11.63 dB to 13.97 dB, giving an improvement of 2.34 dB

Low contrast image



For the last experiment, I reduced the intensity range of the image to make it look flatter and less contrasted. The difference between darker and brighter areas became much smaller, so details in the road, vehicles, trees, and buildings were less noticeable. After applying contrast stretching, the difference between dark & bright areas became much clearer again. This experiment showed the biggest PSNR improvement, increasing from 19.09 dB to 50.86 dB, or 31.77 dB. The improvement is unusually large because the low-contrast image was created using a controlled linear transformation. Since contrast stretching also works by expanding the intensity range, it was able to recover pixel values that were very close to the original image.

Overall PSNR comparison



From the overall comparison, all 4 images had higher PSNR values after enhancement. However, the amount of improvement was different for each problem because every enhancement method works differently & each image was degraded in a different way. The average PSNR increased from 18.43 dB before enhancement to 29.06 dB after enhancement, giving an average improvement of 10.62 dB. In this

experiment, the low contrast case showed the largest PSNR improvement after contrast stretching was applied.

Conclusion

In this assignment, I tested 4 different image problems: blur, excessive brightness, darkness, and low contrast. Each problem needed a different enhancement method, so I used sharpening, gamma correction, logarithmic transformation, and contrast stretching. Overall, all 4 enhanced images showed better PSNR values compared to their degraded versions. The average PSNR increased from 18.43 dB to 29.06 dB, with an average improvement of 10.62 dB. From the results, I learned that image enhancement is not really a “one method fits all thing”. The effectiveness of each method depends on the type of problem and how the image was degraded. Some methods produced a much larger improvement than others, but even the smaller improvements were still useful visually.